

OAC Spark common bean

Raja Khanal, Thomas H. Smith, Thomas E. Michaels, and Karl P. Pauls

Abstract: OAC Spark is a determinate, early maturing navy bean (*Phaseolus vulgaris* L.) cultivar with an upright plant architecture and high yield potential. OAC Spark is adapted to and recommended for the dry bean growing areas in the Prairie provinces of Canada.

Key words: *Phaseolus vulgaris* L., navy bean, early maturity.

Résumé : OAC Spark est un cultivar de haricot (*Phaseolus vulgaris* L.) précoce à croissance déterminée. Il se caractérise par un port droit et un rendement potentiel élevé. On le recommande pour les régions des Prairies canadiennes où l'on cultive le haricot, car la variété y est bien adaptée. [Traduit par la Rédaction]

Mots-clés : *Phaseolus vulgaris* L., haricot, précocité.

Introduction

OAC Spark is an upright navy bean (*Phaseolus vulgaris* L.) cultivar with excellent yield and early maturity with resistance to anthracnose races 73 and 105. OAC Spark was developed at the University of Guelph, Guelph, ON, Canada and was evaluated in the Short Season Narrow Row (SSNR) Co-operative Trials in 2007, 2008, and 2011. OAC Spark was registered by the Variety Registration Office, Canadian Food Inspection Agency, Ottawa, ON, on 9 May 2012 (Registration no. 7204).

Pedigree and Breeding Methods

OAC Spark was developed from a cross between OAC 97-1/OAC 99-1 with the pedigree OAC Speedvale/W34088//OAC Gryphon/W72988. In 1997, the cross between OAC 97-1/OAC 99-1 was designed to improve agronomic traits including plant type with upright growth habit. OAC Speedvale (Seafarer/PI 324685) is an Ontario registered navy bean cultivar with a determinate type I growth habit, early maturity, and good yield (Beattie et al. 2003). Seafarer was released in 1968 as an early-season navy bean variety (Kelly 1999). OAC Gryphon was derived from ExRico 23/Narda followed by 5 backcrosses to ExRico 23. W34088 (Seafarer/Midnight) and W72988 (unknown Belize selection) were used as a source of early maturity.

The population was advanced to the F₅ generation at the Elora Research Station, Ariss, ON using a bulk population advancement approach, where the population was harvested by a combine and bulked to plant the next generation. The F₅ bulk population was planted in the summer of 2001 at the Elora Research Station, Ariss, ON. Single plant selection was performed in the F₅ generation and the resulting F_{5:6} plant-rows were grown in the Elora Research Station in 2002. The main selection criteria for single plants were early maturity, resistance to anthracnose, high yield potential, and desirable plant type with upright architecture. Seeds of the selected single plant were examined for seed size, shape, and colour, and for visual conformity with the standards of the navy bean market class. The lines were further tested and selected in preliminary yield trials at the Elora Research Station, Ariss, ON in 2003. In 2004, the lines were entered into advanced yield trials at the Elora Research Station and a field location near St. Thomas, ON.

Performance

OAC Spark was grown in the Short Season Narrow Row (SSNR) Dry Bean Cooperative Trials as OAC 05-1 in the three Prairie provinces in 2007, 2008, and 2011. The experimental design was a randomized complete block with three replications per location. Test locations with CVs lower than 20% are considered valid tests.

Received 8 February 2016. Accepted 11 April 2016.

R. Khanal, T.H. Smith, and K.P. Pauls. Department of Plant Agriculture, Crop Science Building, University of Guelph, Guelph, ON N1G 2W1, Canada.

T.E. Michaels. Department of Horticultural Science, University of Minnesota, St. Paul, MN 55108, USA.

Corresponding author: Karl P. Pauls (email: ppauls@uoguelph.ca).

Copyright remains with the author(s) or their institution(s). Permission for reuse (free in most cases) can be obtained from [RightsLink](https://www.elsevier.com/locate/RightLink).

Table 1. Yield, days to maturity, and seed weight of OAC Spark navy bean compared with commercial check, Envoy, tested in the Short Season Narrow Row Dry Bean Co-operative Registration Trials during 2007, 2008, and 2011.

Line	Yield (kg ha ⁻¹)				Days to maturity				100 seed weight (g)			
	2007	2008	2011	Mean	2007	2008	2011	Mean	2007	2008	2011	Mean
Envoy	1612	1827	2148	1858	95	97	109	100	14.9	14.8	16.9	15.5
OAC 05-1	1755	1782	2709	2052	94	99	109	101	14.7	12.9	17.8	15.1
No. of trials	3	5	3		3	4	3		3	3	3	
LSD ^a	30	27	53		0.63	0.32	0.83		0.259	0.176	0.15	

^aLSD, least significant difference, based on trial means of cultivars over different locations and across years.

Table 2. Disease reaction and cooking quality and of OAC Spark compared with commercial check cultivar Envoy.

Cultivar	Anthracnose ^a		White mold		Canning quality	
			2008	2011	2011	
	Race 73	Race 105	(1–4) ^b		% hard seed ^c	HC ^d
Envoy	R	S	2	2.2	22.33	1.87
OAC Spark	R	R	2.2	2.5	7.25	1.96

^aAnthracnose reactions were assessed after artificial inoculation under controlled conditions with *C. lindemuthianum* spores (races 73 and 105).

^bWhite mould severity was assessed using a 1 to 4 scale, where 1 = healthy plants, 2 = plants have one infected branch or a pod, 3 = plants have multiple infected branches or pods, and 4 = main stem infected and (or) plants are dead.

^cPercentage hard seed was determined before and after cooking of two hundred seeds soaked in de-ionized water at room temperature for 16 h (overnight) and cooked for 20 min at 97 °C.

^dHydration coefficient (HC) was determined for 500 g dry beans with soaked weight (blanched in 88 °C water for 45 min) divided by dry weight (determined for 500 g of beans).

Agronomic data (i.e., yield, days to maturity, and seed weight (estimated for 100 seeds)) were collected for each plot in each location. Data were subjected to analysis of variance and least square means, standard error of the means, and least significant difference ($P = 0.05$) values were determined. A composite seed sample was formed by mixing approximately 200 g of seed of each entry in each replication from two locations. These samples were processed in the food laboratory at Agriculture and Agri-Food Canada (AAFC), Harrow, ON, and evaluated for cooking and canning quality parameters. Seedling resistance to anthracnose races 73 and 105 of *Colletotrichum lindemuthianum*, and races 1 and 15 of the bean common mosaic virus (BCMV) were tested under controlled conditions in separate growth chambers after inoculation of 10–15 plants at AAFC, Morden, MB. Anthracnose inoculation was done according to the method of Balardin et al. (1997) by brushing both the upper and lower surfaces of fully expanded primary leaves of 7–10 day old seedlings with *C. lindemuthianum* spores (10^6 spores mL⁻¹) cultured in Mathur's medium. OAC 05-1 was also assessed for white mould incidence and severity in an irrigated field disease nursery at AAFC, Lethbridge, AB.

Across 11 location-years in the SSNR Co-operative Trials during 2007, 2008, and 2011, OAC Spark navy bean on average yielded 2052 kg ha⁻¹, which was 10% higher than the check cultivar Envoy (Table 1). OAC Spark matured a

day earlier than the check cultivar Envoy. The seed size of OAC Spark was slightly smaller than the check cultivar Envoy with a seed mass ranging between 14.7 to 17.8 g 100 seed⁻¹. In canning evaluations from seeds from two locations, the hydration coefficient for OAC Spark was similar to the check cultivar Envoy, but it had lower % hard seed compared with the check (Table 2).

OAC Spark is resistant to race 73 and race 105 of *C. lindemuthianum*. OAC Spark carries the SCAR marker SW13 (Melotto et al. 1996) known to be linked to the hypersensitive response gene that conditions resistance to all known races of bean common mosaic virus (BCMV). OAC Spark was similar to the check cultivar Envoy for white mould disease severity (Table 2).

OAC Spark has a determinate growth habit and an upright plant type with good lodging resistance. It has green hypocotyl and white flowers. Plants have medium green leaf colour. The pods are light tan coloured when ripe and covered with short pubescence. Pods have slight to no curvature with a short straight beak. Seeds are white with a dull seed coat lustre and white hilum.

Maintenance and Distribution

OAC Spark was planted in isolation plots for purification and multiplication of seed at Elora Research Station, ON in 2005 and 2006. In 2009, it was planted in a disease-free environment in Idaho for pre-breeder and

breeder seed production. The University of Guelph Bean Breeding Program, Guelph, ON N1G 2W1, Canada will maintain the breeder seed. Pedigreed seed will be distributed by the University of Guelph, Guelph, ON N1G 2W1, Canada, Phone: 519-824-4120, ext. 58339, Fax: 519-763-8933.

Acknowledgements

We acknowledge the financial support provided by the Ontario Bean Growers. The collaboration of Dr. Kirstin Bett and Dr. Parthiba Balasubramanian for conducting the Short Season Narrow Row Dry Bean Co-operative Trials is gratefully acknowledged. We want to thank Mr. Jaret Horner for technical assistance in conducting the Short Season Narrow Row Dry Bean Co-operative Trials.

References

- Balardin, R.S., Jarosz, A.M., and Kelly, J.D. 1997. Virulence and molecular diversity in *Colletotrichum lindemuthianum* from South, Central and North America. *Phytopathology*, **87**: 1184–1191. doi:[10.1094/PHYTO.1997.87.12.1184](https://doi.org/10.1094/PHYTO.1997.87.12.1184). PMID:[18945016](https://pubmed.ncbi.nlm.nih.gov/18945016/).
- Beattie, A.D., Larsen, J., Michaels, T., and Pauls, K.P. 2003. Mapping quantitative trait loci for a common bean (*Phaseolus vulgaris* L.) ideotype. *Genome*, **46**: 411–422. doi:[10.1139/g03-015](https://doi.org/10.1139/g03-015). PMID:[12834057](https://pubmed.ncbi.nlm.nih.gov/12834057/).
- Kelly, J.D. 1999. One hundred years of bean breeding at Michigan State University: a chronology. Mich. Dry Bean Digest. Available from http://agbioresearch.msu.edu/uploads/files/Research_Center/Saginaw_Valley/100YrsMSUBeanBreeding.pdf. Michigan State University, East Lansing, MI.
- Melotto, M., Afanador, L., and Kelly, J.D. 1996. Development of a SCAR marker linked to the I gene in common bean. *Genome*, **39**: 1216–1219. doi:[10.1139/g96-155](https://doi.org/10.1139/g96-155). PMID:[8983191](https://pubmed.ncbi.nlm.nih.gov/8983191/).